Lecture 4: Linear Regression

LSE ME314: Introduction to Data Science and Machine Learning (https://github.com/me314-lse)

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Where Are We?

We left off yesterday with the idea of the **conditional expectation function** (CEF) of Y given some features (e.g. X).

The CEF allows us to recover comparisons of $\mathbb{E}[Y|X]$ across values of X – these comparisons are the heart of many data science applications.

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We also learned about **data generating processes** (DGPs) – probabilistic functions that generate the data we observe.

As we saw yesterday, if we know the DGP, we can trivially recover the CEF.

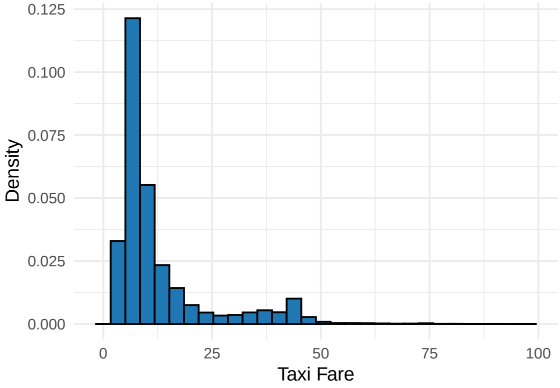
But in practice we don't always know the DGP, so we need to estimate the CEF from observed data based on assumptions about the DGP. Let's take a ride...

Windy City Taxis

Source: John Lee/Chicago Tribune,Chicago Taxi Industry

Windy City Taxis

We're going to work with (a 5% sample of) 1m taxi trips in Chicago, from January 2016. Our outcome variable (aka dependent variable or response variable) will be the fare paid in USD:



Estimands vs. Estimators vs. Estimates

Esti-what?

Three distinct concepts that we will work with today (and for the rest of the course):

- **Estimands**: A target quantity we want to know (e.g., population parameters, or the contrast in two values of the CEF).
- **Estimators**: A function that can take in data and return an estimate of the estimand.
- **Estimates**: The specific values obtained from applying a specific estimator to a specific dataset.

So estimands are our target destination, estimators are our taxi, and estimates are where we actually end up on a given ride?

(Maybe this is not the best analogy.)

The DGP and Estimands

Let's back away from our observed taxi data for a second.

Suppose we assume the following DGP for Y:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where β_0 and β_1 are fixed parameters, and $\epsilon \sim \mathcal{N}(0, 1)$.

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Remember: This is an *assumed* DGP for Y. It's probably wrong in reality.

Now connect this back to the CEF of Y. Given the assumed DGP, our *estimands* are:

- β_0: the intercept, $\mathbb{E}[Y|X = 0]$.
- β_1: the slope, $\frac{\partial \mathbb{E}[Y|X]}{\partial X}$.

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Once we combine any of these with our particular sample, we get an **estimate**.

You can perhaps see that an estimator **is a random variable**, and the estimate is a single draw from that random variable.

Properties of Estimators

Recall the central limit theorem (CLT): Over repeated samples from a random variable, the distribution of sample means is approximately normal.

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It turns out that estimators are themselves random variables – they are functions of the observed data drawn from the probabilistic DGP.

The distribution of an estimator over repeated samples is called its **sampling distribution**. Yesterday's histogram of 1000 simulated dice-roll means was exactly this – the sampling distribution of the sample mean.

The CLT says this distribution is approximately normal for many estimators, not just the sample mean.

Properties of Estimators

We can evaluate estimators in terms of two properties of their sampling distribution:

- **Bias**: The difference between the expected value of the estimator and the true value of the parameter. An estimator is unbiased if its expected value equals the true parameter value.
  - E.g. does $\mathbb{E}[\hat{\mu}] = \mu$? for the mean
  - E.g. does $\mathbb{E}[\hat{\beta}_1] = \beta_1$? for linear regression

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  - E.g. does $\mathbb{E}[\hat{\beta}_1] = \beta_1$? for linear regression

- **Efficiency**: The variability of the estimator across repeated samples:
  - E.g. compare the variance of two proposed estimators: $\text{Var}(\hat{\mu}_1)$ vs. $\text{Var}(\hat{\mu}_2)$

Variance

Yesterday we covered the **expected value** of a random variable – average value of that variable over repeated samples.

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The variance of a random variable (which could be a function!) X is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Read this as: The difference between the random variable and its expected value, squared.

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Intuition check: What is the variance of $\widehat{\mathbb{E}[Y|X]} = 42$?

Zero, because the estimator is a constant so $X = \mathbb{E}[X]$ always.

Bivariate Linear Regression

Assumed Model and Proposed Estimator

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$$Y = \beta_0 + \beta_1 X + \epsilon$$

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Let's estimate these parameters using the ordinary least squares (OLS) linear regression:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

Remember, the function we estimate will be a straight line with some slope $\hat{\beta}_1$ and zero-intercept $\hat{\beta}_0$.

Ordinary Least Squares

The ordinary least squares line is the 'line of best fit'.

This is defined as the line that minimizes:

$$\text{SSR} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

where $\hat{Y}_i$ is the 'predicted value' of Y_i, given the observed value X_i.

SSR stands for **sum of squared residuals**, where $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$ is the residual for i.

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SSR stands for **sum of squared residuals**, where $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$ is the residual for i.

We call this the **objective function** of OLS.

Fitting the linear regression line is thus a minimization problem – there are many possible values of $\hat{\beta}_0$ and $\hat{\beta}_1$, but we need to pick the ones that satisfy this objective function.

Ordinary Least Squares: The Weeds

To see how to solve this minimization problem it's easiest if we work in matrix algebra. This **won't be on the test**!

Assume a $(n \times 1)$ **response vector**, and a $(n \times 2)$ **design matrix** that includes a single feature with an intercept (a column of 1s):

$$\underbrace{\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}}_{\text{Response vector } \mathbf{Y}} \qquad \underbrace{\begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}}_{\text{Design matrix } \mathbf{X}}$$

Ordinary Least Squares: The Weeds

Our estimators can likewise be represented as a vector:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}$$

This is a 2×1 vector, given we want to estimate β_0 and β_1.

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In matrix notation, we can rewrite our objective function as:

$$\hat{\beta} = \arg\min_{\beta} \|Y - X\beta\|^2$$

where $\|Y - X\beta\|^2 = (Y - X\beta)'(Y - X\beta)$.

Ordinary Least Squares: The Weeds

Step 1: Expand the objective function S

$$S(\hat{\beta}) = \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2 = (\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta})$$

$$= \mathbf{Y}'\mathbf{Y} - 2\hat{\beta}'\mathbf{X}'\mathbf{Y} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta}$$

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$$\frac{\partial S}{\partial \hat{\beta}} = -2\mathbf{X}'\mathbf{Y} + 2\mathbf{X}'\mathbf{X}\hat{\beta}$$

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Step 4: Solve for $\hat{\beta}$

$$\begin{aligned} \mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{X}'\mathbf{Y} \\ \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \end{aligned}$$

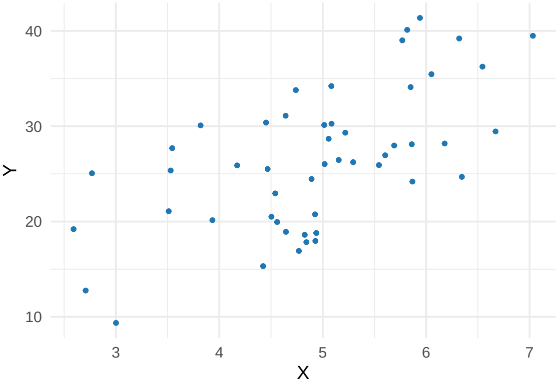
Ordinary Least Squares in Action

Whew!

Ordinary Least Squares in Action

Whew! Let's see how this works in practice:

```r
beta_0 <- 2
beta_1 <- 5
X <- rnorm(50, mean = 5, sd = 1)
Y <- beta_0 + beta_1 * X + runif(50, -10, 10)
```



Ordinary Least Squares in Action

Now, we will solve for the OLS estimates of β_0 and β_1:

```
X_mat <- cbind(1, X)
beta_hat <- solve(t(X_mat) %*% X_mat) %*% t(X_mat) %*% Y
Y_hat <- X_mat %*% beta_hat
residuals <- Y - Y_hat
```

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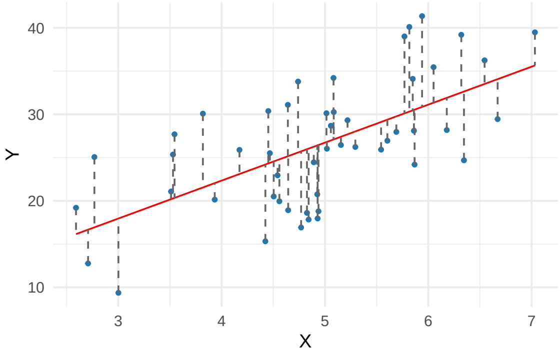
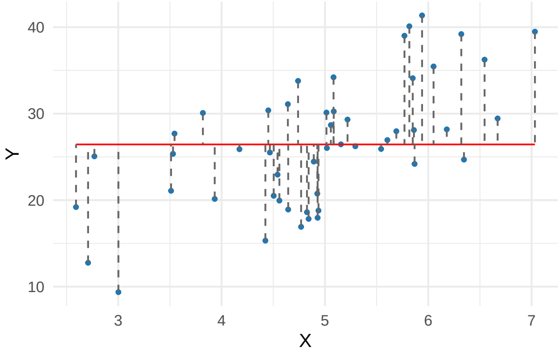


Ordinary Least Squares in Action

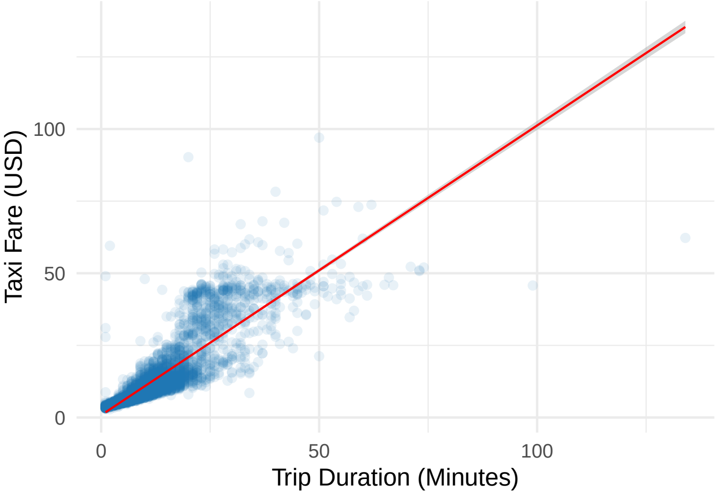
Contrast this performance with the 'null model' – where we explicitly assert that $\beta_1 = 0$:

```
Y_hat_null <- rep(mean(Y), 50)
residuals_null <- Y - Y_hat_null
```



Ordinary Least Squares in Action: Chicago

Let's take this to real data



Ordinary Least Squares in Action

```
lm(fare ~ trip_miles, data = taxi_trips) %>%
  summary() %>%
  broom::tidy()
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   12.5     0.0532      234.        0
2 trip_miles     0.103   0.00233      44.2       0
```

Interpretation:

- Intercept (β_0): The predicted value of Y when $X = 0$
- Slope (β_1): The predicted change in Y for a one unit change in X.

Interpretation: Marginal Changes

The slope gives us the marginal change or marginal effect.

This tells us how Y changes for a one unit change in X.

What this means in a given regression will depend on:

- The units of Y and X.
- The functional form of Y and X (log or not)

OLS Assumptions: Unbiasedness

Under what conditions is our OLS estimator an unbiased estimator of the parameters in the DGP (and thus the population CEF of Y)?

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1. **Linearity**: The true relationship between Y and X is linear *in the parameters*
   - ➜ A violation: the relationship is not linear, e.g., $Y = \beta_0 + {\beta_1}^2 X + \epsilon$.

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   ➜ A violation: the relationship is not linear, e.g., $Y = \beta_0 + {\beta_1}^2 X + \epsilon$.

2. **I.i.d. random sample**: The sample is a random draw.
   ➜ A violation: our sample over-represents certain values from the DGP.

OLS Assumptions: Unbiasedness

3. **No multicollinearity**: Features are not perfectly correlated with each other (this causes non-invertibility of the design matrix $\mathbf{X}$).
   - ➜ A violation: two features that are a linear combination of each other.

OLS Assumptions: Unbiasedness

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4. **Zero conditional mean**: $E[\epsilon|X] = 0$ for all X.
   - ➜ A violation: the error term is correlated with X (e.g. confounding)
   - ➜ Note: This is a big one. It basically means, 'all the systematic components in the DGP are in our estimator'.

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With all four assumptions met, OLS is an unbiased estimator of the parameters in the DGP (and thus the population CEF of Y).

Gauss-Markov Theorem

Consider a fifth assumption:

5. **Homoscedasticity**: $Var(\varepsilon|X) = \sigma^2$ for all X.
   - ➜ The variance of the error term is uncorrelated with the feature X
   - ➜ A violation: Y values in certain parts of the span of X are systematically further from the regression line than others.

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If these five assumptions are met, then we say that linear regression is the **Best Linear Unbiased Estimator** (BLUE) of the (population) CEF of Y. (Best means most efficient here).

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Even when the assumptions are not met, linear regression can still have desirable properties, which is one of the reasons it is the workhorse of econometrics and data science.

Diagnostics: Classical

R^2: The proportion of variance in Y explained by the model.

$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}$

where SSR is the sum of squared residuals (the unexplained variation in Y), and SST is the total sum of squares (all the variation in Y).

Intuition:

- If $R^2 = 0$, then the model explains none of the variation in Y.
- If $R^2 = 1$, then the model explains all of the variation in Y.

Does a high R^2 always make for a good model? (More later)

What Good is a Regression?*

So far we have assumed that we are using regression to estimate the parameters that define the DGP of Y.

In effect, we are using regression as a:

1. *Causal device*:
   - ➜ Estimate the causal effect of a feature on the outcome

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1. *Causal device*:
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But we could use regression in other ways:

2. As a *descriptive device*:
   ➜ Summarize a correlation or co-occurrence between two variables, irrespective of any DGP
3. As a *predictive device*:
   ➜ Fill in 'missing values' (unseen realizations) of Y using X

*Spirling & Stewart (2023)

Multiple Linear Regression

Setup

Let's expand our linear regression model by including more than one feature:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \cdots + \hat{\beta}_k X_k$$

where $\hat{\beta}_0$ is the intercept, $\hat{\beta}_1, \ldots, \hat{\beta}_k$ are the slopes for each feature.

This is called **Multiple linear regression**.

Multiple OLS: Matrix Representation

Note that with matrices we can neatly summarise this case by expanding the design matrix (and the parameter vector):

$$\underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}}_{\text{Estimators } \boldsymbol{\beta}} \qquad \underbrace{\begin{bmatrix} 1 & X_1 & X_2 & \dots & X_k \\ 1 & X_1 & X_2 & \dots & X_k \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_1 & X_2 & \dots & X_k \end{bmatrix}}_{\text{Design matrix } \mathbf{X}}$$

Now we can write down our multiple OLS estimator as:

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Y}$$

Multiple Linear Regression in Action

```
lm(fare ~ trip_mins + trip_miles, data = taxi_trips) %>%
  summary() %>%
  broom::tidy()
```

```
# A tibble: 3 x 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)   1.30     0.0427       30.5 6.29e-202
2 trip_mins     0.946    0.00273     347.  0
3 trip_miles    0.0291   0.00125      23.3 2.01e-119
```

Note that we now have two slopes, one for each feature.

Interpretation

In bivariate regression the slope for any one feature was easy to interpret: the change in Y for a one unit change in that feature.

In multiple regression, we can think of the CEF of Y as an 'n-dimensional hyperplane' (helpful)

The slope for any one feature gives us the change in Y for a one unit change in that feature, holding all other features constant (*ceteris paribus*).

We call this the **partialling out** interpretation, and refer to the slopes as 'partial effects' or 'partial associations'

Interpretation: Partialling Out

What's going on here? Let's compute the partial effect/association of `trip_mins` 'by hand', in two steps.

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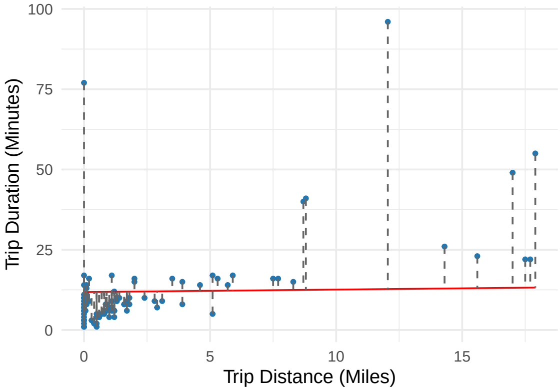
Step 1: Regress `trip_mins` on `trip_miles`, and save the residuals – the variation in `trip_mins` that `trip_miles` cannot explain:

```
# Fit a bivariate regression of mins on miles, and save residuals
lm_mins <- lm(trip_mins ~ trip_miles, data = taxi_trips)
taxi_trips$pred_mins_miles <- predict(lm_mins)
taxi_trips$res_mins_miles <- taxi_trips$trip_mins - taxi_trips$pred_mins_miles
```

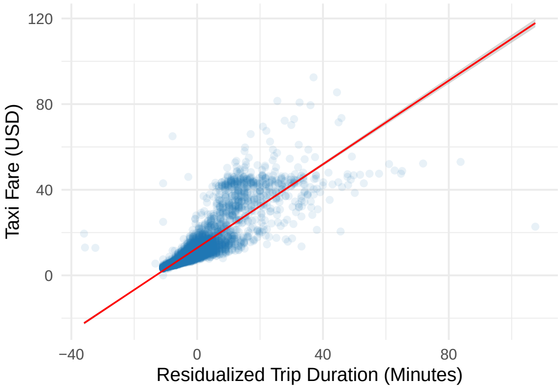
Interpretation: Partialling Out

Each residual is the vertical gap between a trip and the fitted line:



Interpretation: Partialling Out

Step 2: Regress `fare` on those residuals:



Interpretation: Frisch–Waugh–Lovell

```r
# Fit a regression of fare on the residualized mins
lm(fare ~ res_mins_miles, data = taxi_trips) %>%
  summary() %>%
  broom::tidy()
```

```
# A tibble: 2 x 5
  term           estimate std.error statistic p.value
  <chr>             <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)      12.8     0.0298       430.       0
2 res_mins_miles    0.946   0.00292      323.       0
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'Holding all other features constant' thus means using only the variation in `trip_mins` that is left over after `trip_miles` has explained what it can.

More Complex Specifications

We can also include more complex specifications in our regression, such as:

- **Categorical variables**: Qualitative features that we will encode as dummy variables.
- **Polynomials**: Non-linear expansions of features, such as squares or cubes.
- **Interactions**: Flexibility that allows features to affect Y differently as a function of other features.

More Complex Specifications: Categorical Variables

```
# What are the levels of the categorical variable?
unique(taxi_trips$payment_type)
```

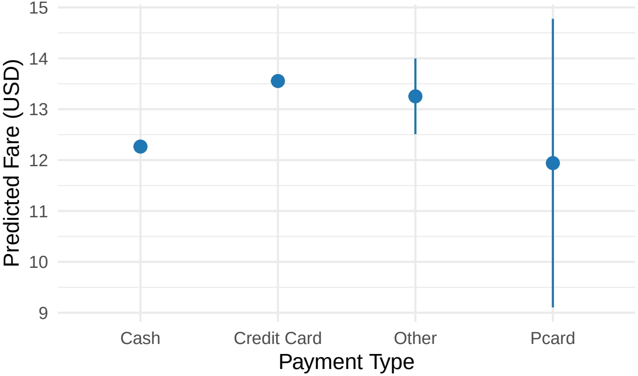
```
[1] "Cash"        "Credit Card" "Other"       "Pcard"
```

```
lm(fare ~ trip_mins + trip_miles + factor(payment_type),
   data = taxi_trips) %>%
  summary() %>%
  broom::tidy()
```

```
# A tibble: 6 x 5
  term                              estimate std.error statistic   p.value
  <chr>                                <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)                         0.850    0.0469     18.1   4.55e- 73
2 trip_mins                           0.937    0.00273   343.    0
3 trip_miles                          0.0288   0.00124    23.2   1.31e-118
4 factor(payment_type)Credit Card    1.29     0.0566     22.8   3.86e-114
5 factor(payment_type)Other           0.987    0.381       2.59  9.63e-  3
6 factor(payment_type)Pcard          -0.325    1.45       -0.224 8.22e-  1
```

More Complex Specifications: Categorical Variables

Here we visualize the predicted fare paid based on different payment types, holding `trip_mins` and `trip_miles` constant at their means.



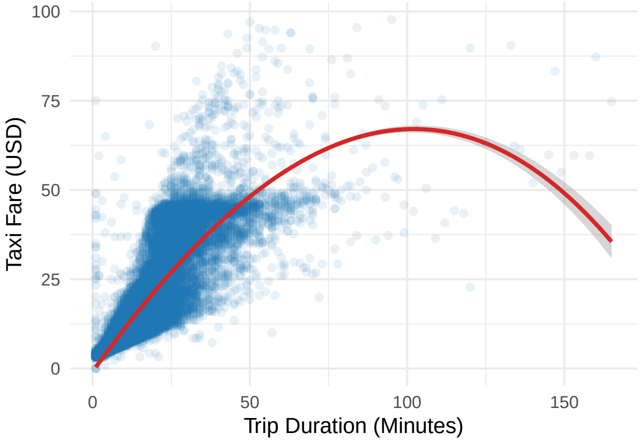
More Complex Specifications: Polynomials

```r
lm(fare ~ trip_mins + I(trip_mins^2) + I(trip_mins^3),
   data = taxi_trips) %>%
  summary() %>%
  broom::tidy()
```

```
# A tibble: 4 x 5
  term            estimate  std.error statistic  p.value
  <chr>              <dbl>      <dbl>     <dbl>    <dbl>
1 (Intercept)    -0.851      0.0662        -12.9  8.71e-38
2 trip_mins       1.24       0.00897       138.   0
3 I(trip_mins^2) -0.00468    0.000259      -18.1  1.05e-72
4 I(trip_mins^3) -0.00000893 0.00000163     -5.48 4.24e- 8
```

More Complex Specifications: Polynomials

Visualising what's going on:



More Complex Specifications: Interactions

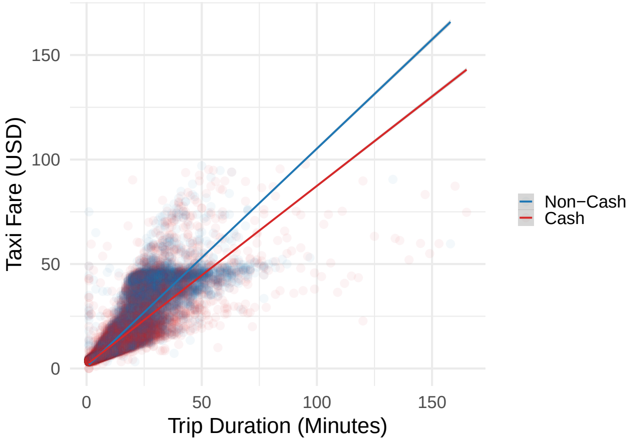
```
lm(fare ~ trip_mins * cash_dummy,
   data = taxi_trips) %>%
  summary() %>%
  broom::tidy()
```

```
# A tibble: 4 x 5
  term                estimate std.error statistic   p.value
  <chr>                  <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)            0.789   0.0670       11.8 5.43e- 32
2 trip_mins              1.04    0.00383     273.  0
3 cash_dummy             1.03    0.0864       11.9 1.47e- 32
4 trip_mins:cash_dummy  -0.189   0.00535     -35.3 5.79e-269
```

Interpretation: `fare` increases as `trip_mins` increases (marginal effect $= 1.04$), but less so (marginal effect $= 1.04 - 0.189 = 0.851$) when the trip is paid for in cash rather than alternatives.

More Complex Specifications: Interactions

Visually:



Inference

One Regression, One Estimate

Today we have run several regressions, and each time we got a single number for each $\hat{\beta}$.

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No. Recall that estimators are random variables. Our $\hat{\beta}_1$ is **one draw** from its sampling distribution.

Inference is about characterising that distribution, and using it to quantify the uncertainty in our estimates.

Simulating the Sampling Distribution

Let's see this in practice. Suppose we treat our taxi data as the **population** of trips, and then draw repeated samples from it:

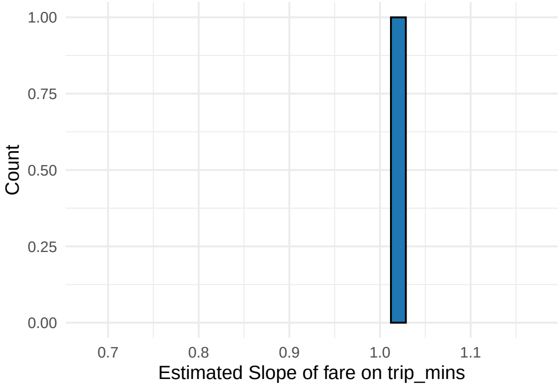
```r
beta_sampler <- function(K, n) {
  beta_hats <- numeric(K)
  for (k in 1:K) {
    # Draw one sample of n trips from the 'population'
    trips_sample <- taxi_trips[sample(nrow(taxi_trips), n), ]
    # Fit the regression on this sample, store the slope estimate
    trips_lm <- lm(fare ~ trip_mins, data = trips_sample)
    beta_hats[k] <- coef(trips_lm)["trip_mins"]
  }
  return(beta_hats)
}

many_beta_hats <- beta_sampler(K = 1000, n = 500)

# Set aside one sample of the same size -- 'our' sample for later
taxi_sample <- taxi_trips[sample(nrow(taxi_trips), 500), ]
```

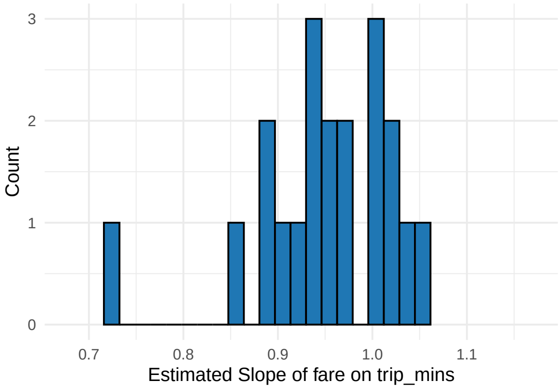
Simulating the Sampling Distribution

One sample gives us one $\hat{\beta}_1$:



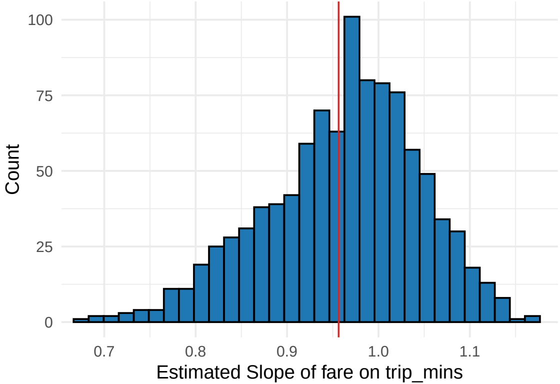
Simulating the Sampling Distribution

Twenty samples give us twenty $\hat{\beta}_1$'s:



Simulating the Sampling Distribution

And 1000 samples give us this, thanks to the **Central Limit Theorem**:



This is (approaching) the **sampling distribution** of $\hat{\beta}_1$ – centred on the population slope (red line), with some **spread**.

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Recall variance – the spread of a random variable. Its square root is the **standard deviation** – the same spread on the original scale.

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The standard deviation of the sampling distribution of an estimator is called the **standard error**:

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Read this as: How far our estimates tend to fall from the centre of the sampling distribution.

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The problem: we never see the sampling distribution – we get one sample, and one $\hat{\beta}_1$. So we have to **estimate** the standard error.

Two Approaches to Inference

We need to characterise the sampling distribution of $\hat{\beta}_1$ using only the one sample we have. Let's look at two approaches:

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1. **Classical asymptotic inference**: Estimate the standard error with a closed-form formula, and rely on the CLT for the shape of the distribution.
2. **The bootstrap**: Reconstruct the sampling distribution empirically, by re-sampling our own data.

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We need to characterise the sampling distribution of $\hat{\beta}_1$ using only the one sample we have. Let's look at two approaches:

1. **Classical asymptotic inference**: Estimate the standard error with a closed-form formula, and rely on the CLT for the shape of the distribution.

2. **The bootstrap**: Reconstruct the sampling distribution empirically, by re-sampling our own data.

We start with the classical approach, and take it through hypothesis tests, *p*-values, and confidence intervals. Then we return to the bootstrap.

Classical Asymptotic Inference

Define a **null hypothesis** (H_0) and an **alternative hypothesis** (H_A).

H_0: Under this hypothesis, we believe that the true value of the parameter is zero ($\beta_1 = 0$)

H_A: Under this hypothesis, we believe that the true value of the parameter is something other than zero ($\beta_1 \neq 0$)

We ask: How (un)likely is the data we observe, if we assume the H_0 is true?

Classical Asymptotic Inference

How do we do this?

1. Calculate a test-statistic of interest (often a t-statistic or F-statistic)
2. Assume an asymptotic distribution for the test-statistic (t-distribution or F-distribution) **under the null hypothesis** (drawing on the CLT)
3. Compare the observed test-statistic to this reference distribution
4. Calculate a p-value, which is the probability of observing a test-statistic as extreme as the one we observed, **given that the null hypothesis is true**.

Classical Asymptotic Inference

To test linear regression coefficients we use the t-statistic:

$$t = \frac{\hat{\beta}_1 - \beta_{H0}}{\mathrm{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\mathrm{se}(\hat{\beta}_1)}$$

where β_{H0} is the value of the parameter under the null hypothesis, $\hat{\beta}_1$ is the estimated coefficient, and $\mathrm{se}(\hat{\beta}_1)$ is the standard error of the estimate.

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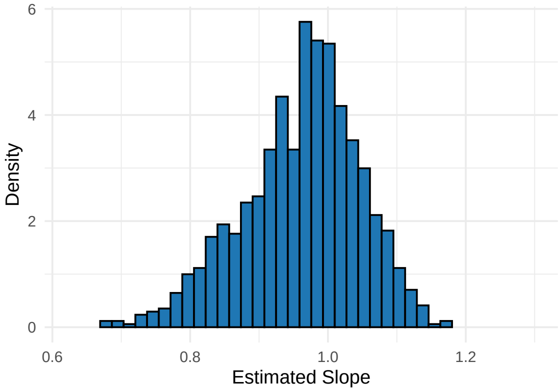
What is the t-statistic? Ratio of the magnitude of the estimate to the implicit variability of the estimate.

Under H_0, the t-statistic follows a known reference distribution – the t-distribution, which approaches the standard normal as n grows. Where does that come from? Not from our data...

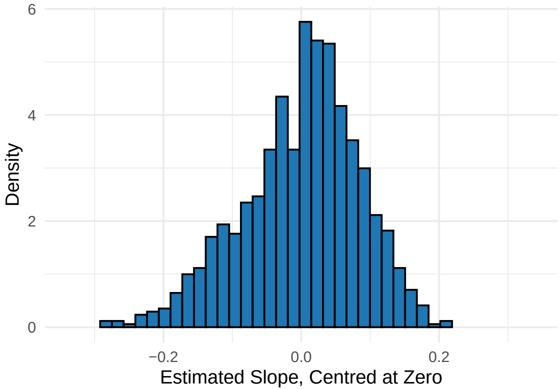
The Null Distribution

Start from the sampling distribution of $\hat{\beta}_1$ that we simulated:



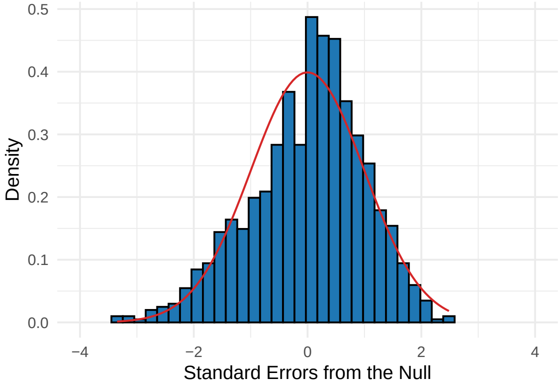
The Null Distribution

Under H_0 the true slope is zero, so re-centre the distribution at zero:



The Null Distribution

Then divide by the standard error. What remains is approximately $\mathcal{N}(0, 1)$:



This is the reference distribution we test against – it comes from the CLT, not from our data.

Classical Asymptotic Inference in Action

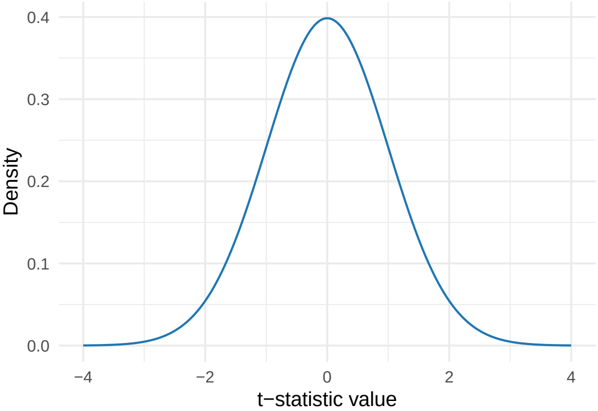
Calculating the t-statistic for our polynomial regression:

```r
# To make the t-stat more plausible, take a subsample:
set.seed(5)
taxi_sub <- taxi_trips %>%
  sample_frac(0.005)

# Fit the cubic polynomial regression, extract the t-stat on trip_mins
t_stat <- lm(fare ~ trip_mins + I(trip_mins^2) + I(trip_mins^3),
             data = taxi_sub) %>%
  summary() %>%
  broom::tidy() %>%
  mutate(t_stat = estimate / std.error) %>%
  filter(term == "trip_mins") %>%
  select(t_stat)

t_stat
```

```
# A tibble: 1 x 1
  t_stat
   <dbl>
1   2.54
```

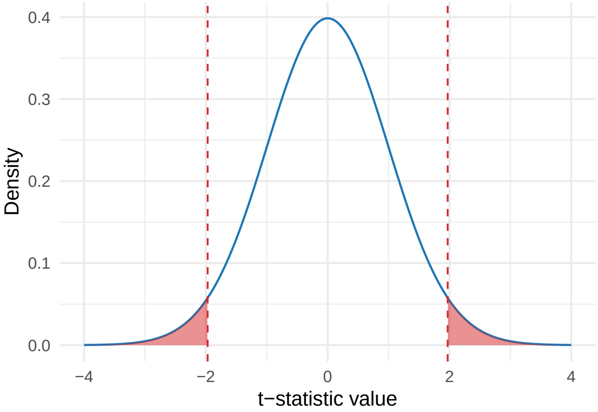
Classical Asymptotic Inference in Action

Start with the null distribution of the t-statistic:



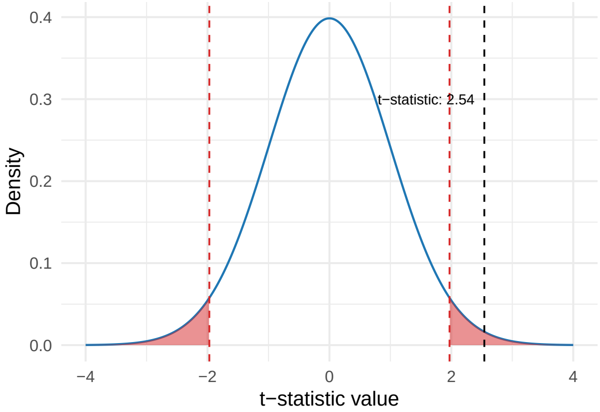
Classical Asymptotic Inference in Action

Add the **rejection region**: if the observed t-statistic lands in the shaded tails, we reject H_0 at $\alpha = 0.05$:



Classical Asymptotic Inference in Action

Now place our observed t-statistic:



If it falls in the shaded region, we reject H_0.

The p-value

The ***p*-value** is:

$$p = \Pr(|t| \geq |t_{\text{obs}}| \mid H_0)$$

Read this as: If the null hypothesis were true, how likely is a test-statistic at least as extreme as the one we observed?

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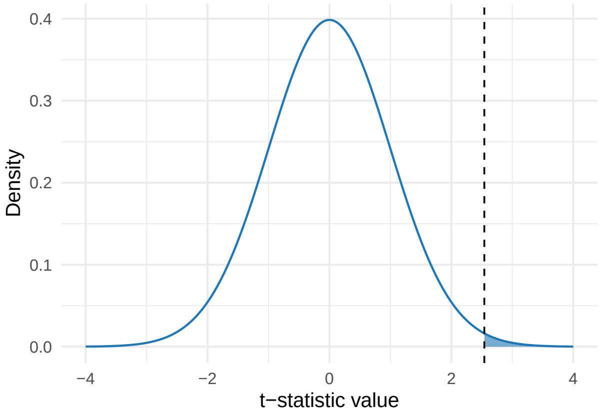
Read this as: If the null hypothesis were true, how likely is a test-statistic at least as extreme as the one we observed?

A common misreading: The p-value is the probability that H_0 is true. No – H_0 is either true or false; it is not a random variable.

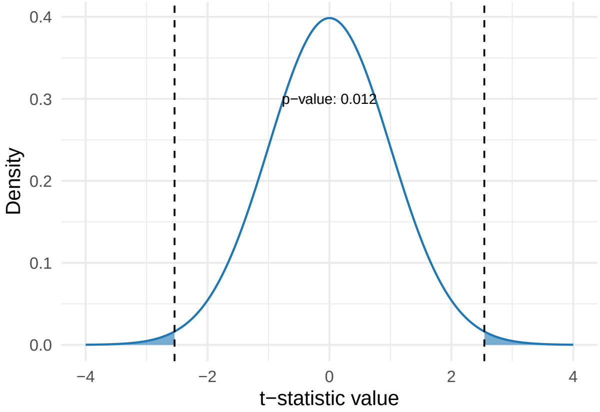
The p-value in Action

Mark our observed t-statistic, and shade the tail at least as extreme in its direction:



The p-value in Action

Our alternative is two-sided, so add the same area in the other direction. The shaded area *is* the p-value:



The p-value falls below α exactly when the t-statistic falls beyond the critical values – the two views always agree.

Confidence Intervals

We can use the standard error to construct a **confidence interval** for the parameter of interest, against the reference distribution (e.g. the t-distribution):

$$\hat{\beta}_1 \pm t_{crit} \cdot \mathrm{se}(\hat{\beta}_1)$$

where t_{crit} is the critical value from the t-distribution at the desired confidence level (e.g. 95%, corresponding to $\alpha = 0.05$).

Interpretation: Over repeated samples, 95% of the 95% confidence intervals we construct will contain the true value of the parameter.

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A common misreading: 'there is a 95% chance that β_1 is in this interval.' No – the parameter is fixed, it is the interval that varies over repeated samples.

Bringing it All Together

If the *p*-value is small, it is unlikely that our test-statistic is drawn from the null distribution.

Thus we might feel inclined to 'reject the null hypothesis' at some α level.

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At $\alpha = 0.05$, the following are the **same test**:

- t-test: reject H_0 if $|t_{\text{obs}}| > t_{crit} \approx 1.96$
- p-value: reject H_0 if $p < 0.05$
- Confidence interval: reject H_0 if the 95% CI excludes zero

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The three will always agree as they are three views of the same object: the sampling distribution of $\hat{\beta}_1$.

Heteroskedasticity-Robust Inference

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In some cases, like the linear probability model, it is by definition violated.

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In some cases, like the linear probability model, it is by definition violated.

We can adjust for this by using **heteroskedasticity-robust standard errors**. Essentially, we allow each observation to have its own error variance, rather than assuming constant variance across observations.

There are various implementations in `R`:

- `{sandwich}::vcovHC()` used with `lm()`
- `{estimatr}::lm_robust()`
- `{fixest}::feols()`
- `{modelsummary}::modelsummary()`

General advice: Always estimate heteroskedasticity-robust standard errors.

Cluster-Robust Inference

Recall also that we had hierarchies in some of our datasets.

We might want to adjust for the fact that the residuals within each cluster are correlated. For example: trips by the same driver, or trips on the same day, may have correlated residuals. We can do this by using **cluster-robust standard errors**.

Essentially, we allow our errors to be correlated within clusters, relaxing the i.i.d assumption.

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- `{fixest}`::`feols()`
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When to cluster? Cluster at the level at which your sampling was clustered, or at which your feature of interest varies – not on every grouping you can find.

The Bootstrap

Intuition: The best approximation we have of the population **is our sample**. So: **re-sample the sample**, with replacement, to mimic drawing new samples from the population.

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The **nonparametric bootstrap** works as follows:

- B times:
  1. Draw a random sample X_b^* of size n from the original sample **with replacement**
  2. Estimate the chosen model
  3. Store the chosen test statistic (e.g. $\hat{\beta}_b^*$) from the fitted model.

With $\hat{\beta}^* = \{\hat{\beta}_1^*, \ldots, \hat{\beta}_B^*\}$ in hand we can:

- Estimate the se of $\hat{\beta}$ as the standard deviation of $\hat{\beta}^*$
- Compute 95% confidence intervals using the 2.5th and 97.5th percentiles of $\hat{\beta}^*$

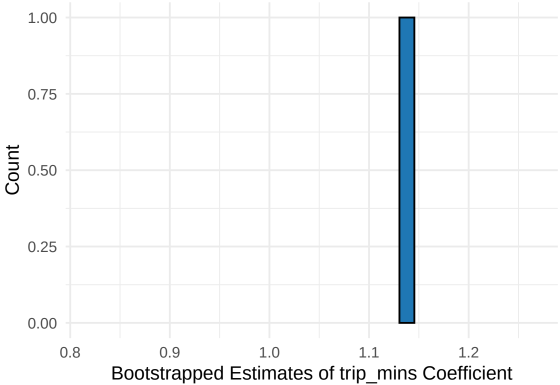
The Bootstrap in Action

```r
# Pretend taxi_sample -- set aside earlier, n = 500 -- is all our data
boot <- function(sims){
  # Create a vector to store the estimates
  estimates <- numeric(sims)

  for (i in 1:sims) {
    # Sample with replacement
    sample_data <- taxi_sample[sample(nrow(taxi_sample), replace = TRUE), ]
    # Fit the model
    model <- lm(fare ~ trip_mins, data = sample_data)
    # Store the estimate of the coefficient
    estimates[i] <- coef(model)["trip_mins"]
  }

  return(estimates)
}

boot_estimates <- boot(sims = 500)

# Compare: bootstrap SE vs. sd of the simulated sampling distribution
c(bootstrap = sd(boot_estimates), sampling_dist = sd(many_beta_hats))
```

```
 bootstrap sampling_dist
0.07566485    0.08463933
```

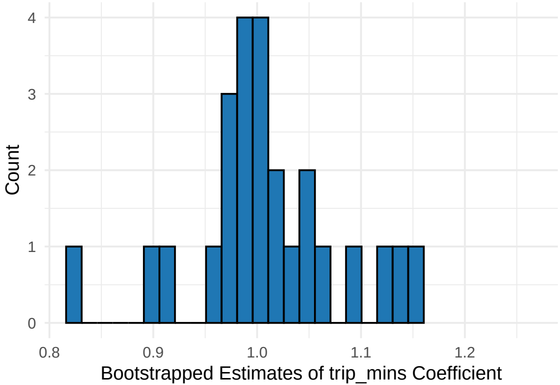
The Bootstrap in Action

One re-sample gives us one $\hat{\beta}_1^*$:



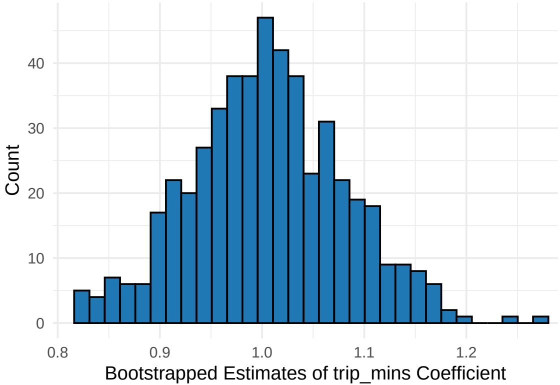
The Bootstrap in Action

Twenty-five re-samples give us twenty-five $\hat{\beta}_1^*$'s:



The Bootstrap in Action

And all 500:



This **empirical distribution** is our bootstrapped proxy of the sampling distribution of the estimator $\hat{\beta}_1$.

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Classical (robust) standard errors: Default to these, as they are computationally fast, transparent, well understood, and appropriate whenever the estimator has a known sampling distribution (e.g. OLS).

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Either way, the target is the same object – the sampling distribution of the estimator.